

Project Design Phase
Solution Architecture

Date	29-06-2026
Team ID	23A51A05H9
Project Name	Leave Approval System
Maximum Marks	4 Marks

Solution Architecture:

The **Leave Approval System** is built on the ServiceNow platform using a layered architecture approach. Users interact with the system through a custom application form that captures request parameters such as leave type, start date, end date, and manager. The application logic layer instantly validates the date ranges on the client-side to prevent entry errors and utilizes Workflow Studio (Flow Designer) to execute the background business routing, automate manager approvals, and orchestrate real-time system state updates.

- **User Interaction Layer:** A custom form layout provides a streamlined, self-service interface where employees can easily select their leave types, specify dates, define their approving manager, and provide business justifications.
- **Business Logic Layer:** Client-side scripts intercept form interactions to validate input criteria (automatically preventing end dates from being set earlier than start dates), while a cloud-native Flow Designer workflow automates the transactional logic, manager approval gates, and data-field state changes.
- **Data & Service Layer:** A secure ServiceNow scoped table stores the primary leave request records and transaction fields, while system logs, multi-channel automated emails, and native UI badges handle outbound delivery, stakeholder communication, and audit trail maintenance.

Solution Architecture Diagram:

